BATTERY HACKATHON

# Rebuilding the invisible:
## consumption & feed-in from residual load profiles.

*A challenge for anyone who wants to size commercial batteries on the data customers actually send — not the data we wish they had.*

# Why this data problem matters

*Sizing a battery for self-consumption optimisation is a 15-minute problem. If you can't tell when power went into the grid, you can't tell what a battery would have charged.*

## Greenfield — no PV yet

Customer sends a clean consumption load profile.

**Battery simulation is straightforward:**

Simulate PV production, subtract self-consumed share, cycle a battery through the surplus, evaluate savings.

## Brownfield — PV already installed

Customer sends a residual load profile (Netzbezugslastgang).

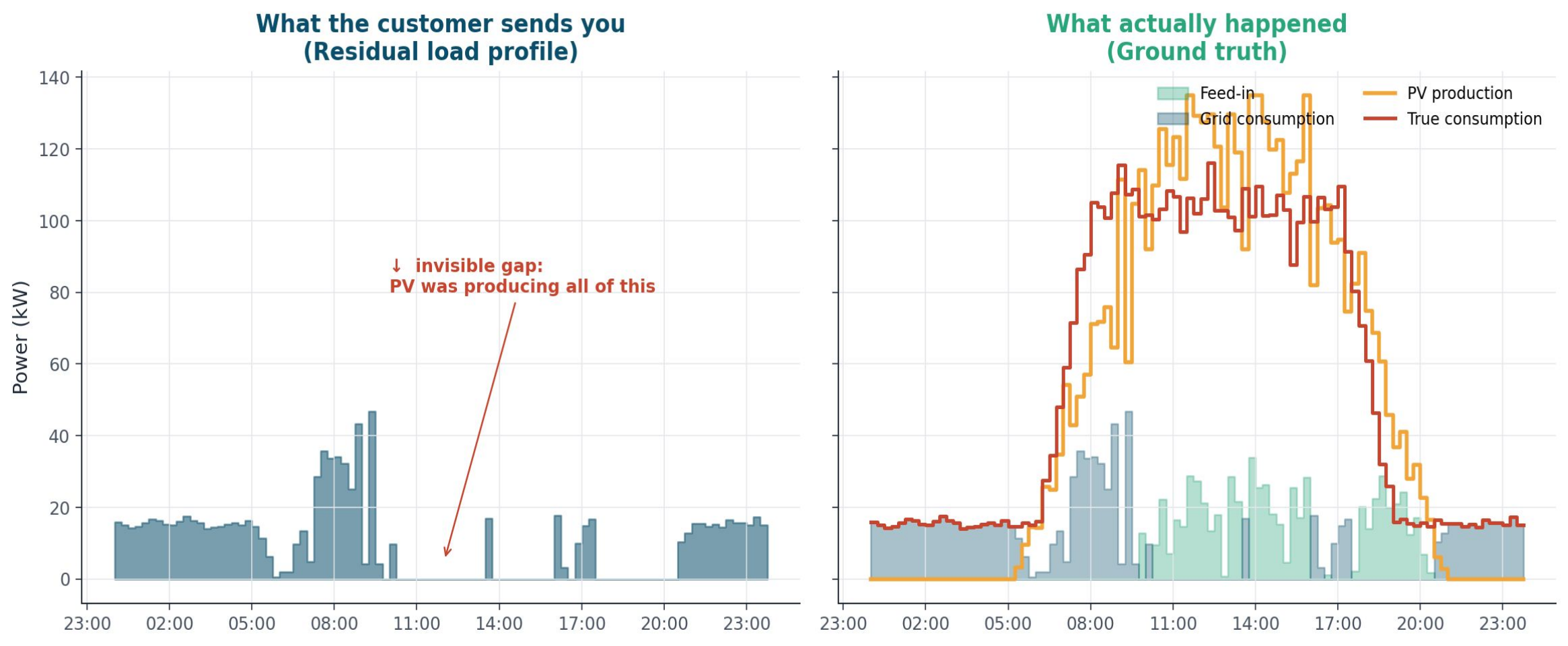
**Self-consumption already happened in the meter — and it is invisible to you.**

Intervals with surplus PV now show 0 instead of a negative value. You can no longer tell when or how much energy went to the grid.

*Every real commercial inquiry we see is brownfield.*

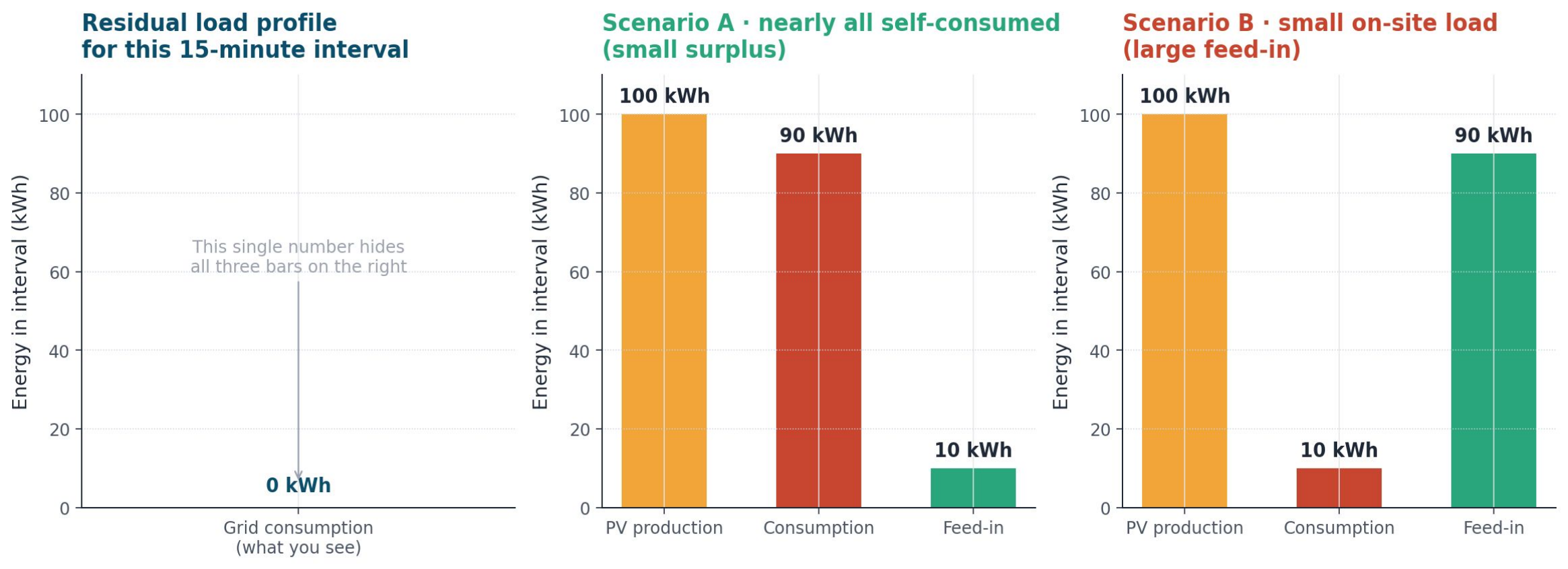
# One summer day, two very different views

*Left: what arrives in your inbox.  Right: what actually happened behind the meter.*



# A single 0 hides several very different realities

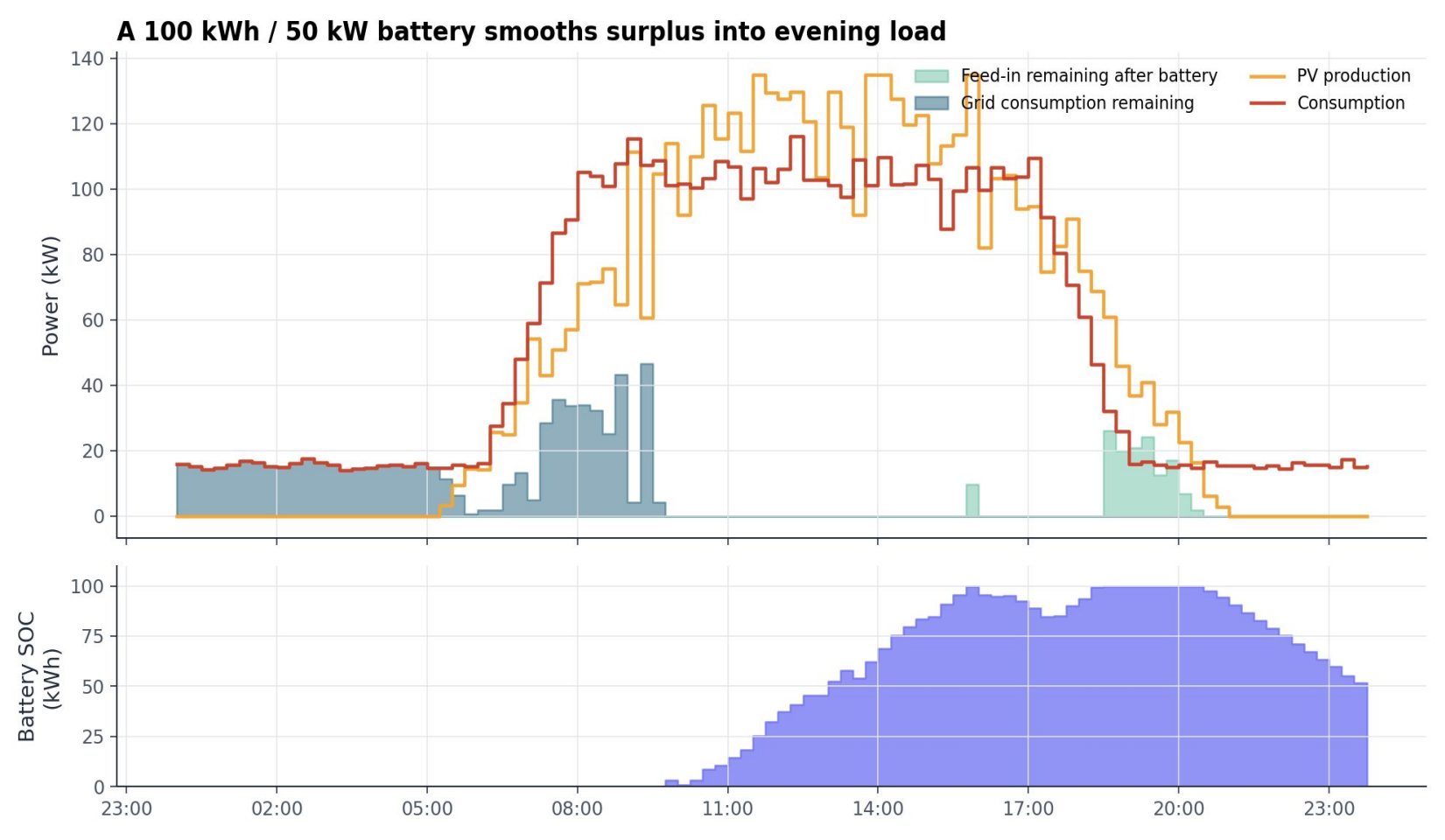
*When grid consumption reads 0, the only thing you can say for sure is: production ≥ consumption. Everything else is your assumption.*




**Same residual value.** Opposite implications for a battery: Scenario A barely benefits from storage; Scenario B leaves 90 kWh on the table every interval.

# What a battery simulation actually needs

*A self-consumption-optimised battery lives in the gap between PV surplus and evening load. That gap is exactly what a residual profile hides.*




## Battery inputs it cannot fake

**When surplus starts and ends**

→ determines charge windows

**How much surplus in each 15-min slot**

→ caps the charge rate and daily harvest

**When unmet demand occurs**

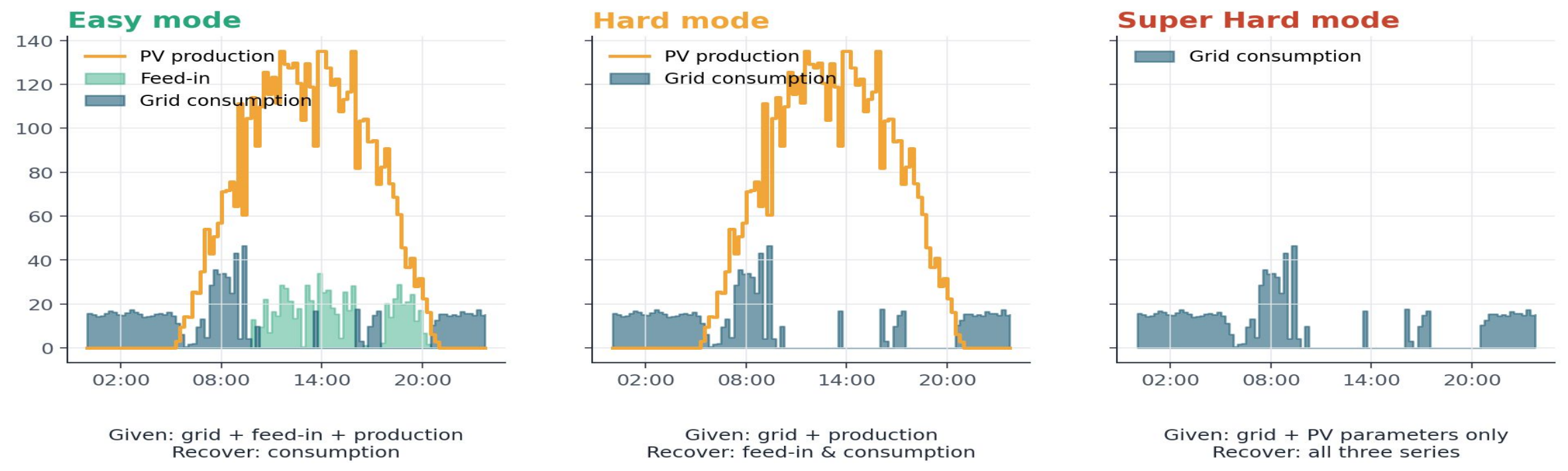
→ determines discharge windows

**Seasonal & weekly shape**

→ holidays and weekends change the cycling profile

# Three challenge modes

*Pick your own difficulty. All three use the same synthetic year of 15-minute data for a Munich C&I consumer with a 150 kWp PV system.*




## Easy

- Given: grid charge + feed-in
- Recover: original consumption via PV-Production-Simulation

## Hard

- Given: grid + production + annual SC ratio
- Recover: feed-in & consumption
- Exploit: constraint that grid>0 ⇒ feed-in=0

## Super Hard

- Given: grid + PV system parameters only
- Recover: production, feed-in, consumption
- Bring: weather reanalysis + a simulator you trust

# Come build with us.

*The datasets are synthetic, the problem is painfully real.*

01
**Pick a mode**

02
**Find patterns**

03
**Test on a new file**